

TrendMart International

Global E-Commerce Platform Migration

Executive Summary

FINAL EXAM USE-CASE

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REASON FOR CRITICAL TRANSFORMATION:

TrendMart international is facing fast decline in their business due to the usage of outdated infrastructure. Currently, it is documented that the company is losing around \$8 Million dollars annually due to their inefficient infrastructure. They often face system failures during the peak shopping seasons as this further leads to 40% of customers to abandon their purchases. Further, they faced 6-hour black Friday crash which even decreases the customer's trust in the TrendMart e-commerce service. Moreover, many customers reported that they face slower page loads due their shopping sessions. In addition, TrendMart is also facing compliance issues as they in a position to pay €1.2M fine for data protection violations which happened last year. Finally, they still use 200 physical servers to run their single large application and still use 30TB traditional database which shows performance issues. All these issues occur due to their 5-year-old infrastructure cannot handle modern e-commerce demands. Thus, the need to adopt to AWS based cloud computing infrastructure in order to make their business stay relevant with modern e-commerce infrastructure and to compete effectively with their competitors. This makes the cloud migration an urgent strategic priority for TrendMart International.

HIGH LEVEL SOLUTION APPROACH:

In order to mitigate the above-mentioned issues, the proposed solution will utilize the distributed AWS architecture for building a scalable, resilience infrastructure and even a compliance following one. In this solution, TrendMart will transition its current monolith structure to a microservices where the services deployed widely across three AWS regions, mainly in North America, Europe, and Asia. Here, they offer low-latency global access with the help of AWS Cloudfront for sub-millisecond load times, AWS EKS/ECS for scalable microservices. Also, utilizing the Aurora Database and DynamoDB for high-performance data storage. Further, utilizing SQS/SNS for handling the traffic smoothly during peak loads. Thus, with this architecture, we can ensure TrendMart to have 99.99% uptime while following strict EU data residency and with the ability to support 1,000,000 users during peak shopping events along with automatic scaling feature.

EXPECTED BENEFITS AND RETURN ON INVESTMENT:

With this transformation, we aim to reduce significant operational risks while improving customer experience and financial performance. Moreover, the infrastructure cost is projected to reduce from \$450K/month to under \$300K/month with the effective utilizing of AWS cloud computing services. As a result of this transformation, TrendMart can eliminate the outages and speed up

their services. Most importantly, they restore millions in lost sales and boost their conversion rates. Again, it gives TrendMart a chance to rebuild their customer trust with faster checkouts and consistent user experience. Also, this gives a better chance to introduce new features faster that will improve the competitiveness and guide us to a long-term growth. Finally, this transformation is expected to yield us over \$10 Million in annual financial benefit.

HIGH-LEVEL TIMELINE AND MAJOR MILESTONES:

The overall expected time frame for the completion of this transformation is around 12-18 months. During the initial months, the primary focus will be on establishing a secure AWS Landing Zone with identity governance and covering the compliance baselines. Then, it moves on to focusing on deploying the container platform and building CI/CD pipelines to support new microservices. Further, the focus will be on migrating the 30TB legacy database to Aurora for near-zero downtime and utilizing the AWS Data migration services for it. Later on, services like product catalog, user management, inventory, cart and payment services are gradually built. Finally, the overall cost optimization, operational automation and the complete withdrawal of on-prem infrastructure will be taking place. With this, the TrendMart's cloud-first transformation will be completed and is ready for new era for ecommerce.